Introduction

In today's digital world, cyber security has become a major concern due to the increasing number of cyber-attacks such as hacking, phishing, and unauthorized access. Traditional security systems are not always efficient in detecting new and complex threats. Therefore,

This project focuses on developing a web-based application that can analyze user behavior and detect suspicious activities. The system uses machine learning algorithms to identify patterns in data and classify activities as safe or threats.

In addition to this, the proposed system enhances security by continuously monitoring user activities and identifying unusual patterns in real time. By analyzing parameters such as login frequency, IP address changes, and user behavior, the system can quickly detect anomalies that may indicate potential cyber threats. This reduces the dependency on manual monitoring and helps in faster decision-making.

Furthermore, the integration of a web-based interface using HTML, CSS, and JavaScript makes the system easy to use and accessible. The backend developed in Python ensures efficient processing and accurate predictions using machine learning models. Overall, this project demonstrates how combining AI with modern web technologies can create an effective and scalable solution for cyber security threat detection.

Objective of the Study

The main objectives of this project are:

- To develop a system that can detect cyber threats using machine learning.
- To analyze user activities such as login attempts and IP changes.
- To provide real-time alerts for suspicious activities.
- To create a user-friendly web interface using HTML, CSS, and JavaScript.
- To improve accuracy and efficiency in threat detection.

Project Category

Web Based Application

Flask is a lightweight and flexible Python web framework used to build web applications and APIs efficiently. In this project, Flask is used to connect the machine learning model with the frontend interface, enabling real-time cyber threat detection. The system analyzes user activity and provides instant results through a web-based dashboard.

Hardware Requirement:-

This software requires the following minimum hardware configuration:
Processor: Pentium-IV or above
RAM: 4 GB or above
Hard Disk: 128 GB minimum
Internet Connection (for testing APIs and datasets)

Software Requirement:-

Windows / Linux Operating System
Python (3.x version)
XAMPP Server (optional for database)
VS Code / Sublime Text
Web Browser (Chrome/Edge)

Language Used

HTML
CSS
JavaScript
Python

Technologies and Framework Used :-

Flask (Backend Framework)
Scikit-learn (Machine Learning)
Pandas (Data Processing)
NumPy (Numerical Operations)

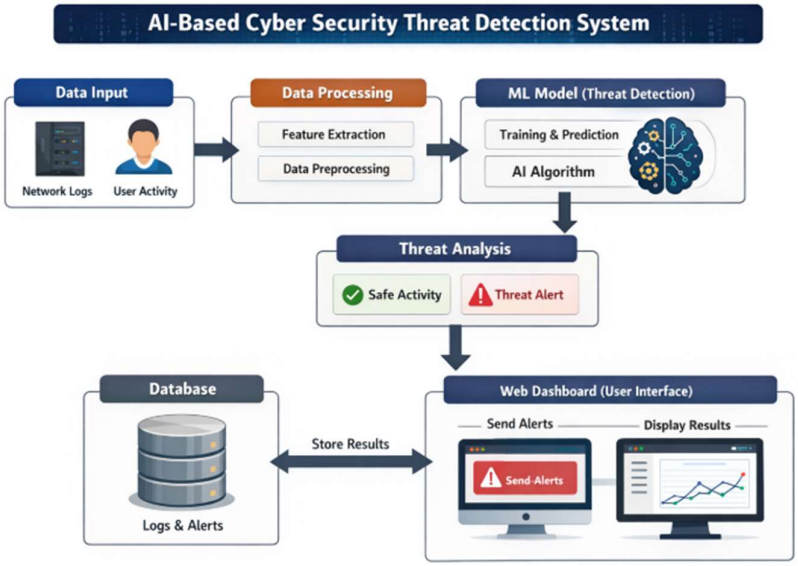
SOFTWARE REQUIREMENT ANALYSIS

Server Side Technology :-
Python, Flask, Machine Learning (Scikit-learn)

Client Side Technology :-
HTML, CSS, JavaScript

Database Server :-
MySQL

Complete Structure of the System



SYSTEM MODULES

Secure Login / Logout

- User authentication system
- Secure login with username & password
- Logout option for security

Dashboard

- View system overview
- Display total users, threats detected
- Real-time monitoring of activities
- Alerts and notifications panel

Threat Detection Module

- Analyze user activity (login attempts, IP changes)
- Detect suspicious behavior using ML model
- Classify activity as Safe or Threat
- Generate instant alerts

Data Management (Insert / Delete / View / Update / Search)

- Insert new activity records
- Delete unwanted data
- View all logs
- Update existing records
- Search records by user/IP/date

Add New Record

- Add new user activity data manually
- Store logs for analysis

Activity Logs

- Maintain history of user actions
- Store detected threats
- Track system performance

Reports Module

- Filter data by date range
- Search by user name / IP address
- View all detected threats
- Generate monthly/yearly reports

Admin Panel

- Manage all users
- View all records
- Monitor system activity
- Control alerts and logs

Profile Management

- Update user details
- Change password
- Manage account settings

Reference

Python Software Foundation. *Python Official Documentation.*
Available at: https://docs.python.org/3/

Scikit-learn Developers. *Scikit-learn: Machine Learning in Python Documentation.*
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Pallets Projects. *Flask Web Framework Documentation.*
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NumPy Developers. *NumPy Documentation.*
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Date :-

Signature of the students

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